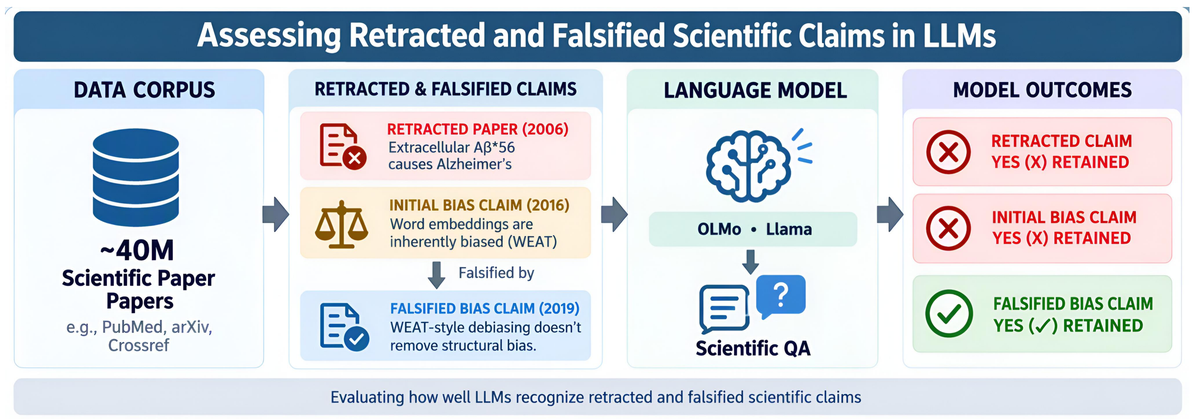
Can Scientific Claims Be Removed from Large Language Models? A Systematic Evaluation of Claim-Level Unlearning

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MOTIVATION



PROBLEM FORMULATION

Given a model M_θ, and dataset $D = \{F_1, F_2, R_{ext}, R_{int}\}$, where, $(F_1 \cup F_2) \cap (R_{ext} \cup R_{int}) = \emptyset$, an unlearning algorithm A updates the model parameters to get θ', the objective is defined as -

$$\min_{\theta'} \underbrace{\mathbb{E}_{(q,a)\in F_i}[l_f(a|q;\theta')]}_{\text{Forget}} + \lambda \underbrace{\mathbb{E}_{(q,a)\in R}[l_r(a|q;\theta')]}_{\text{Retain}}$$

Where, (q, a) is the QA pair and λ balances forgetting and retention loss.

Objective (Validation of Claim Unlearning):
Model unlearned on F_1 -
F_1 - Forgotten ✔ F_2 - Measure Claim Level Forgetting
Model unlearned on F_2 -
F_1 - Measure Claim Level Forgetting F_2 - Forgotten ✔

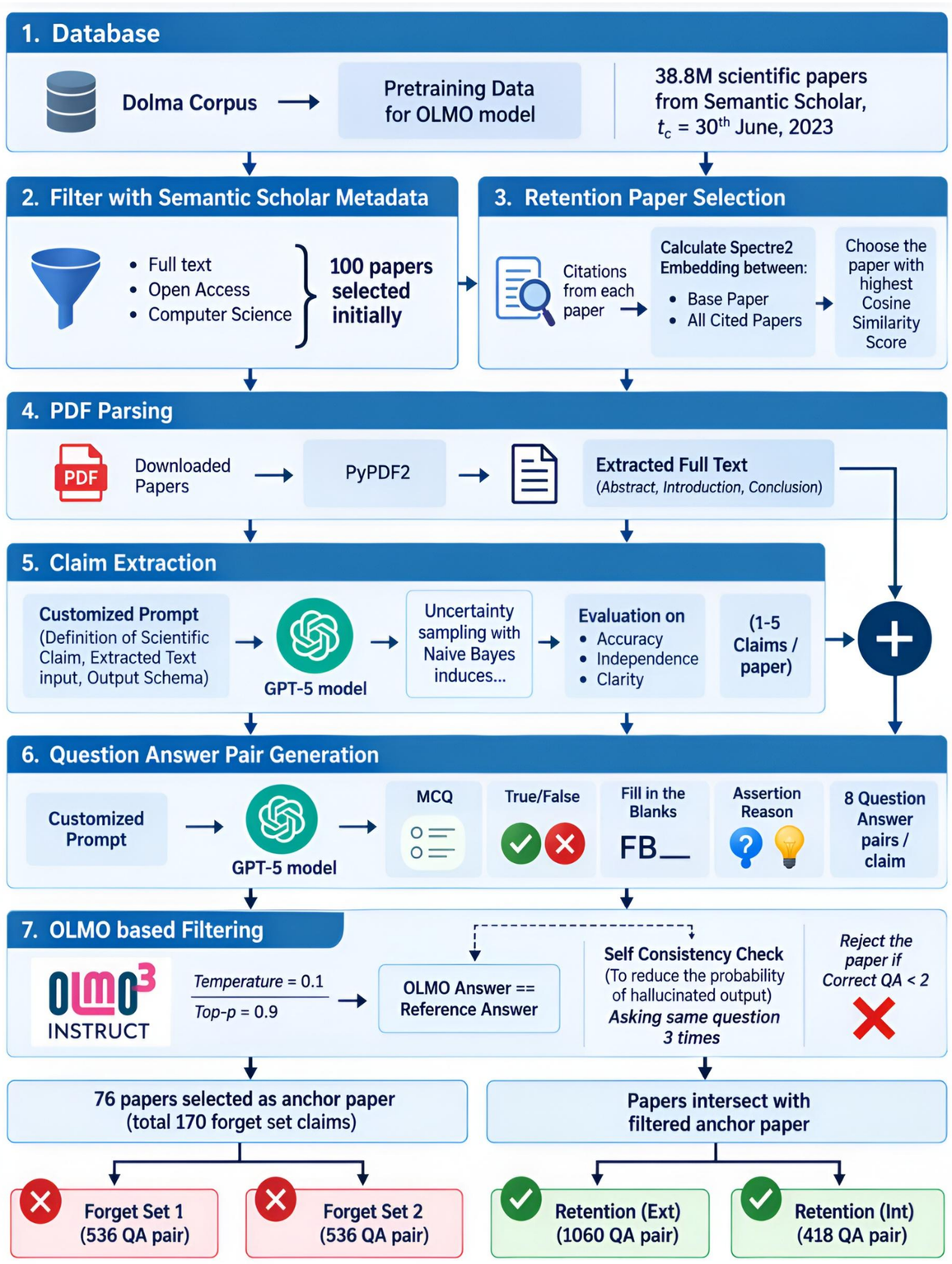
EVALUATION PROTOCOL

Forget: exact match (EM) and ROUGE-L F1 on F_1 and F_2.
Retain: unrelated information from cited papers and source papers.
General utility: MMLU, ARC-Challenge and HellaSwag.
MIA: compare direct and semantic members with unrelated non-members.
Min-K% (k = 20%) AUC near 0.5 indicates indistinguishability.
Larger RefScore (L_unlearned − L_base) indicates stronger forgetting.

ALGORITHMS USED

Gradient Difference **(GD)**, Negative Preference Optimization **(NPO)**, Simplified NPO **(SimNPO)**, NPO with retain set **(NPO+RT)**, SimNPO with retain set **(SimNPO+RT)**.

DATASET CREATION



Dataset	Papers	Claims	F_1	F_2	R_{ext}	R_{int}
Computer science	76	170	536	536	1060	418
Medical	78	127	508	508	908	543
Retracted	91	141	390	390	884	671

RESULTS AND DISCUSSION

Model	Method	Forget Set 1 ↓		Forget Set 2 ↓		Retain (Ext) ↑		Retain (Int) ↑		General Benchmarks		
		R-F1	EM	R-F1	EM	R-F1	EM	R-F1	EM	MMLU	Arc-C	HellaSwag
OLMo	Base Model	97.54	94.21	93.92	89.17	95.40	90.75	92.44	83.49	58.75	54.35	76.02
	GD (F_1)	**71.70**	**58.20**	93.51	88.24	**93.66**	84.15	90.53	72.00	56.44	50.08	70.13
	NPO+RT (F_1)	88.81	81.52	93.67	89.55	92.44	**86.13**	90.66	**82.05**	**58.28**	49.91	69.89
	GD (F_2)	91.65	78.35	89.60	83.58	93.27	84.52	91.02	77.03	57.54	49.57	70.93
	NPO+RT (F_2)	96.14	89.92	**86.78**	**82.27**	91.47	85.18	**91.60**	81.81	57.39	**50.51**	**73.02**
LLaMA	Base Model	96.61	93.65	87.37	77.61	91.86	86.03	90.07	83.25	64.74	75.58	55.88
	GD (F_1)	78.33	74.06	88.32	80.03	85.45	80.37	89.00	83.25	63.01	73.91	46.67
	NPO+RT (F_1)	**73.56**	**70.33**	90.91	86.38	86.53	**83.01**	**90.35**	**83.73**	63.35	**75.28**	53.41
	GD (F_2)	93.31	88.80	**59.18**	**37.87**	76.78	65.47	81.31	73.20	60.93	74.30	47.69
	NPO+RT (F_2)	96.08	93.09	78.59	66.23	**87.36**	78.86	85.14	77.27	**64.44**	75.01	**55.37**

Table 1. LORA based unlearning results for OLMo and LLaMA. (Only on top performing algorithms)

Model	Method	Forget Set 1 ↓		Forget Set 2 ↓		Retain (Ext) ↑		Retain (Int) ↑		General Benchmarks		
		R-F1	EM	R-F1	EM	R-F1	EM	R-F1	EM	MMLU	Arc-C	HellaSwag
OLMo	Base Model	92.16	78.59	84.59	65.96	87.74	71.14	92.49	81.95	58.76	54.35	76.02
	GD (F_1)	**58.20**	**35.78**	75.40	61.05	86.78	73.78	92.77	84.34	57.24	52.47	71.07
	NPO+RT (F_1)	83.09	67.36	83.93	68.07	89.69	74.77	92.49	85.63	**59.50**	52.73	**75.30**
	GD (F_2)	83.80	68.77	**55.59**	**32.28**	88.97	74.77	93.95	86.37	56.43	51.70	72.50
	NPO+RT (F_2)	92.52	85.64	73.89	81.53	**91.96**	**85.06**	**94.79**	**88.07**	59.14	**53.41**	73.54
LLaMA	Base Model	86.23	71.92	76.64	56.49	80.39	63.32	90.24	82.68	64.75	75.58	55.88
	GD (F_1)	70.08	55.78	73.70	56.14	81.61	65.96	93.81	88.21	**63.87**	**75.39**	**52.99**
	NPO+RT (F_1)	**41.70**	**26.66**	41.70	15.43	87.18	70.15	**97.83**	**94.47**	47.29	74.97	54.18
	GD (F_2)	83.32	68.77	68.34	48.77	82.94	67.62	93.90	88.39	63.10	75.29	52.82
	NPO+RT (F_2)	45.98	30.18	**32.79**	**7.37**	**88.86**	**79.07**	96.06	93.00	43.20	74.48	49.57

Table 2. Full parameter unlearning results for OLMo and LLaMA. (Only on top performing algorithms)

HUMAN VALIDATION

Experts reviewed **75 claims** from **50 papers**. Human–LLM agreement on quality criteria reached Gwet's AC2 of **0.84–0.91**.

Accuracy: Claims and answers match the source evidence.

Independence: Each item is self-contained and needs no unstated context.

Clarity: Wording is clear and unambiguous.

KEY TAKEAWAYS

Can LLMs truly forget scientific claims?

Existing methods do not achieve true claim level forgetting, only some papers not deeply rooted in the pretraining corpus can transfer the unlearning behaviour to the complementary forget set.

What else is needed for true Claim Level Unlearning?

To read the full paper scan the QR code here: